

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1686

Cement Plant 138 kV
September 03, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1686. The Interconnection Customer proposes a 228.1 MW solar photovoltaic facility interconnecting at the Cement Plant 138 kV substation in the Virginia area. The request was received on March 04, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 493 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,706,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: HYAK to DUPLESSIS 138 kV
Contingency: CEMENT PLANT to 43RD & S. LAND 138 kV
Loading: 140.2% (Rating: 243 MVA)
DFAX: 52.8%
Pre-project loading: 90.6%
2. Facility: US ORD DEPOT-HERLONG to DUPLESSIS 138 kV
Contingency: CEMENT PLANT to 43RD & S. LAND 138 kV line
Loading: 133.9% (Rating: 249 MVA)
DFAX: 52.8%
Pre-project loading: 85.6%
3. Facility: CEMENT PLANT to 43RD & S. LAND 138 kV
Contingency: TOLSON to US ORD DEPOT-HERLONG SUBSTATION 138 kV
Loading: 128.5% (Rating: 366 MVA)
DFAX: 67.2%
Pre-project loading: 86.6%
4. Facility: OROVADA to TOLSON 138 kV
Contingency: CEMENT PLANT to 43RD & S. LAND 138 kV L
Loading: 121.2% (Rating: 353 MVA)
DFAX: 52.8%
Pre-project loading: 87.0%
5. Facility: TOLSON to US ORD DEPOT-HERLONG 138 kV
Contingency: CEMENT PLANT to 43RD & S. LAND 138 kV Line
Loading: 119.8% (Rating: 367 MVA)
DFAX: 52.8%
Pre-project loading: 87.0%

6. Facility: 43RD & S. LAND 345/138 kV Autotransformer
Contingency: TOLSON to US ORD DEPOT-HERLONG 138 kV L
Loading: 117.2% (Rating: 401 MVA)
DFAX: 67.2%
Pre-project loading: 79.0%
7. Facility: PALMER to NORTH BRISTOL 138 kV Line
Contingency: WOLF HILLS ENERGY to E BERKSHIRE 345 kV L
Loading: 108.6% (Rating: 342 MVA)
DFAX: 43.8%
Pre-project loading: 79.3%
8. Facility: PALMER to WOLF ILLS ENERGY 138 kV L
Contingency: E BERKSHIRE to WOLF HILLS ENERGY 345 kV line
Loading: 107.5% (Rating: 360 MVA)
DFAX: 43.8%
Pre-project loading: 79.7%
9. Facility: NASSAU to LITCHFIELD 138 kV L
Contingency: LOST NATION to NO TROY 138 kV
Loading: 104.7% (Rating: 192 MVA)
DFAX: 11.7%
Pre-project loading: 90.9%
10. Facility: SPRINGBANK SUBSTATION to LITCHFIELD 138 kV
Contingency: LOST NATION to NO TROY 138 kV
Loading: 104.3% (Rating: 214 MVA)
DFAX: 11.7%
Pre-project loading: 91.8%
11. Facility: GORGE DICKIE to BROADFORD 138 kV
Contingency: KENTUCKY DAM to STOCK ISLAND 138 kV line
Loading: 102.5% (Rating: 174 MVA)
DFAX: 10.7%
Pre-project loading: 88.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HYAK to DUPLESSIS 138 kV	Rebuild line with higher-capacity conductor	\$3,033,000	22% (\$664,000)
US ORD DEPOT-HERLONG to DUPLESSIS 138 kV	Rebuild line with higher-capacity conductor	\$6,516,000	22% (\$1,426,000)
CEMENT PLANT to 43RD & S. LAND 138 kV	Rebuild line with higher-capacity conductor	\$8,659,000	24% (\$2,052,000)
OROVADA to TOLSON 138 kV	Rebuild line with higher-capacity conductor	\$13,963,000	20% (\$2,776,000)
TOLSON to US ORD DEPOT-HERLONG 138 kV	Rebuild line with higher-capacity conductor	\$9,051,000	20% (\$1,800,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
43RD & S. LAND 345 kV/138 kV Auto	Replace transformer with higher-rated unit	\$18,148,000	24% (\$4,300,000)
PALMER to NORTH BRISTOL 138 kV Line	Reconductor line	\$1,553,000	29% (\$445,000)
WOLF HILLS ENERGY to PALMER SUBSTATION 138 kV	Reconductor line	\$782,000	29% (\$224,000)
NASSAU to LITCHFIELD 138 kV L	Reconductor line	\$2,449,000	13% (\$312,000)
SPRINGBANK SUBSTATION to LITCHFIELD 138 kV	Reconductor line	\$3,232,000	13% (\$411,000)
GORGE DICKIE to BROADFORD 138 kV	Reconductor line	\$3,015,000	10% (\$296,000)
	Total allocated to project		\$14,706,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.